

Master on Statistics for Data Science
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Multivariate Analysis

“Dataset description and PCA. ”

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1. INTRODUCTION

In many real-world scenarios, data does not consist of a single measurement but of multiple, related observations. In multivariate analysis, each observation is viewed as a random vector, $\mathbf{X} = (X_1, X_2, \dots, X_p)'$, where p random variables represent different measurements on a single subject. A key challenge, and the central focus of multivariate analysis, is that these variables are typically intercorrelated. Unlike univariate methods, which examine variables in isolation, multivariate procedures are designed to “make allowance for these correlations,” providing a deeper and more accurate understanding of the underlying structures in the data.

This project will apply the principles of multivariate analysis to a practical, complex dataset: *Encuesta sobre el Gasto de los Hogares en Educación (EGHE) for the 2023/2024 academic year*, published by the Spanish National Statistics Institute (INE). This official survey provides a rich source of data, detailing not only household expenditure on education but also a wide range of associated socioeconomic, demographic, and educational variables.

The primary objective of this study is to employ descriptive multivariate procedures to analyze this dataset. Once we have reduced the original dataset to a subset that meets the requirements that need to be met for this project, we will begin with a thorough data description that allows us to understand in depth how the variables are distributed. Next, we will apply Principal Component Analysis (PCA), a technique that will allow us to disentangle the overlapping information in the correlated variables by constructing a small number of uncorrelated linear combinations (principal components) that reveal the essential dimensionality of the system. Through these methods, we aim to uncover the complex, multidimensional patterns that govern educational expenditure in Spain.

2. DATASET

The data used in this study comes from the *Encuesta sobre el Gasto de los Hogares en Educación (EGHE)* for the academic year 2023/2024, published by the Spanish National Statistics Institute (INE). This official survey provides detailed information on household expenditure related to education, covering a wide range of socioeconomic, demographic, and educational variables.

2.1. Classification of the Variables

The original dataset contains 81 variables that describe different dimensions of educational spending, household characteristics, and schooling conditions in Spain. From this extensive set of variables, we selected a subset of eleven variables that we considered the most relevant for our multivariate analysis and that met the required conditions for this project (at least 6 quantitative, 2 binary, and 2 multiclass categorical). In particular, the selected variables include numerical measures of educational expenditure, binary indicators capturing individual educational status and access to financial aid, and categorical descriptors such as nationality and type of schooling.

This curated selection allows us to focus on the most meaningful dimensions of educational expenditure while maintaining enough variability to perform a comprehensive multivariate analysis.

In particular, our variables can be classified regarding their nature as follows:

■ Qualitative variables:

- **Nominal:** ‘sex’, ‘nationality’, ‘student’, ‘help’, ‘type’.
- **Ordinal:** There are no ordinal variables because there is no order relationship between the categories.

■ Quantitative variables:

- **Discrete:** There are no discrete variables.
- **Continuous:** ‘age’, ‘uniform’, ‘food’, ‘books’, ‘computer’, ‘activities’.

However, for this project, it will be more convenient to classify the variables into three types: numeric, binary, and categorical. A more detailed description of this classification is given in the following tables.

2.1.1. Numeric Variables

Variable	Description	Possible Values / Range
Age	Age	Positive values
Uniform	Expenditure on school uniforms, shoes, and related accessories (euros)	Positive values
Food	Expenditure on school dining services (euros)	Positive values
Books	Expenditure on textbooks and supplementary study resources (euros)	Positive values
Computer	Expenditure on IT products (euros)	Positive values
Activities	Expenditure on extracurricular activities (euros)	Positive values

2.1.2. Binary Variables

Variable	Description	Possible Values / Range
Sex	Sex	0 = Male, 1 = Female
Student	Whether the individual is undertaking any studies	0 = No, 1 = Yes
Help	Whether the individual receives any educational grants or subsidies	0 = No, 1 = Yes

2.1.3. Categorical variables

Variable	Description	Possible Values / Range
Nationality	Nationality	1 = Spanish, 2 = Foreign, 3 = Double nationality
Type	Type of schooling	1 = Public Schooling, 2 = State-Funded Private Schooling, 3 = Private Schooling

2.2. Data Cleaning and Pre-processing

After downloading the original dataset from the INE and selecting the variables, the next step was to reduce the number of observations from 10211 to a number between 100 and 1000.

First, given that the aim of the selected variables in our dataset was to describe the expenditure related to education, we decided that a sensible decision was to only consider those individuals who were students. Thus, we selected the subset of the dataset corresponding to students and deleted the 'student' variable since it was no longer useful, obtaining a dataset with 10 variables and 5061 observations.

Next, we examined the amount and distribution of missing values (NAs) across all variables, and we discovered that several variables presented extremely high proportions of missing data—some, like ‘uniform’, ‘food’, ‘computer’, and ‘activities’ reached percentages of NAs greater than 67 %. In these cases, imputation would not have been meaningful, since the available information was too scarce to generate reliable estimates. Thus, we decided to approach NAs in a different way, and we analyzed the missing values at the observation level. It was seen that several observations contained multiple NAs across different variables, therefore carrying substantially less information compared to other observations with less or non-missing variable values. Following this reasoning, we decided to remove observations that presented several missing values.

Once this filtering criterion was applied, it was seen that several variables still contained a substantial proportion of missing values. In particular, out of a total of 374 observations, ‘computer’ had 222 missing values, books had 53, and ‘activities’ had 41. Thus, we decided to implement Multiple Imputation to fill the remaining NAs in our data. Simple imputation methods would distort the data and underestimate variability, especially with missingness of this magnitude. In contrast, Multiple Imputation—implemented with the *mice* package using Predictive Mean Matching—generates realistic values that preserve the multivariate structure of the data and properly reflect uncertainty, thereby minimising bias and retaining as much information as possible.

The resulting dataset has a total of 374 observations and 10 variables, and the methods employed for obtaining it ensure that the subsequent multivariate analyses are based on consistent, reliable, and sufficiently informative data. The statistical programming language used for performing the described preliminary data processing was *R*.

2.3. Exploratory Data Analysis (EDA)

Before proceeding to any modeling, we first conduct a brief Exploratory Data Analysis (EDA) using the statistical programming language *R*. The goal of this preliminary stage is to gain an initial understanding of the dataset by examining the structure of the variables and identifying basic patterns. This step provides essential context for the subsequent analysis and helps guide methodological decisions.

We are going to perform a brief Exploratory Data Analysis (EDA) using the statistical programming language *R*.

2.3.1. Binary and Categorical Variables

```
> summary(df[,c("sex", "nationality", "help", "type")])
```

sex	nationality	help	type
male :382	Spanish :757	no :603	public :445
female:429	Foreign : 35	yes:208	funded-private:317
	2 nationalies: 19		private : 49

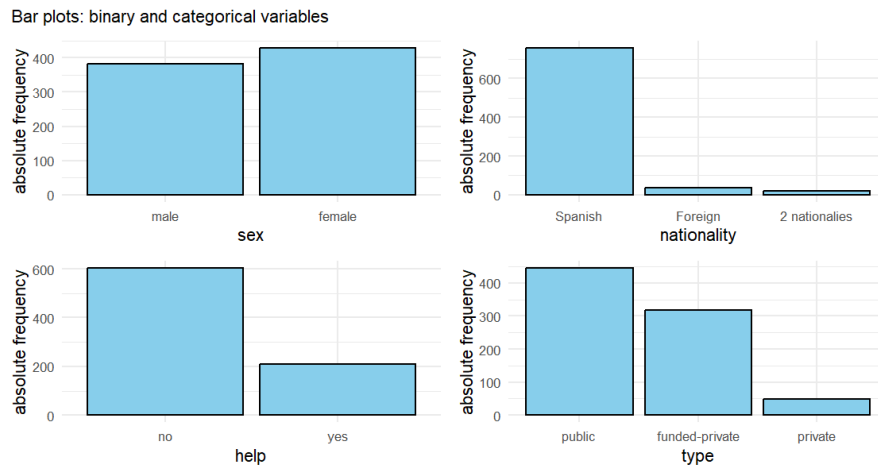


Fig. 2.1. Barplots of the binary and categorical variables EDA

Based on the bar plots of Figure 2.1 and the displayed summary statistics, we can describe the binary and categorical variables. The 'sex' variable is fairly balanced, with a slight majority of females (429) compared to males (382). In contrast, the binary 'help' variable shows a significant imbalance, with the 'no' category (603) being almost three times more frequent than the 'yes' category (208). Looking at the multi-category variables, 'type' is distributed across three groups: 'public' is the most common category with 445 observations, followed by 'funded-private' at 317, and then by 'private', which is a small minority, accounting for only 49 observations. Finally, the 'nationality' variable exhibits the most extreme skew. It is overwhelmingly composed of 'Spanish' nationals (757), while the 'Foreign' (35) and '2 nationalities' (19) categories are very small.

2.3.2. Continuous Variables

```
> summary(df[,c("age", "uniform", "food", "books", "computer", "activities")])
```

age	uniform	food	books	computer	activities
Min. : 3.000	Min. : 10.0	Min. : 0.0	Min. : 0.00	Min. : 10.0	Min. : 0.0
1st Qu.: 7.000	1st Qu.: 60.0	1st Qu.: 600.0	1st Qu.: 70.25	1st Qu.: 133.8	1st Qu.: 200.0
Median :10.000	Median :100.0	Median : 995.0	Median : 163.00	Median : 250.0	Median : 330.0
Mean : 9.783	Mean :149.5	Mean : 943.2	Mean : 191.76	Mean : 388.5	Mean : 405.1
3rd Qu.:12.000	3rd Qu.:200.0	3rd Qu.:1300.0	3rd Qu.: 281.50	3rd Qu.: 589.0	3rd Qu.: 517.5
Max. :53.000	Max. :800.0	Max. :3000.0	Max. :1040.00	Max. :1800.0	Max. :3250.0

The histograms seen in Figure 2.2, along with the displayed summary, provide a visual overview of the distribution of the continuous variables in the dataset, revealing distinct patterns for each. The 'age' variable is mostly concentrated between 3 and 15 years, with a few older observations extending up to 53, consistent with the median of 10 years. Expenditure variables such as 'uniform', 'books', 'computer', and 'activities' show marked right-skewed distributions, with the majority of observations near lower expenditure values but with some extreme values reaching up to 800 for 'uniform', 1040 for 'books', 1800 for 'computer', and over 3200 for 'activities'. The 'food' variable presents a more spread distribution with a clear peak between 600 and 1300. These patterns indicate subs-

tantial variability in spending, with a small proportion of higher expenditures affecting the means.

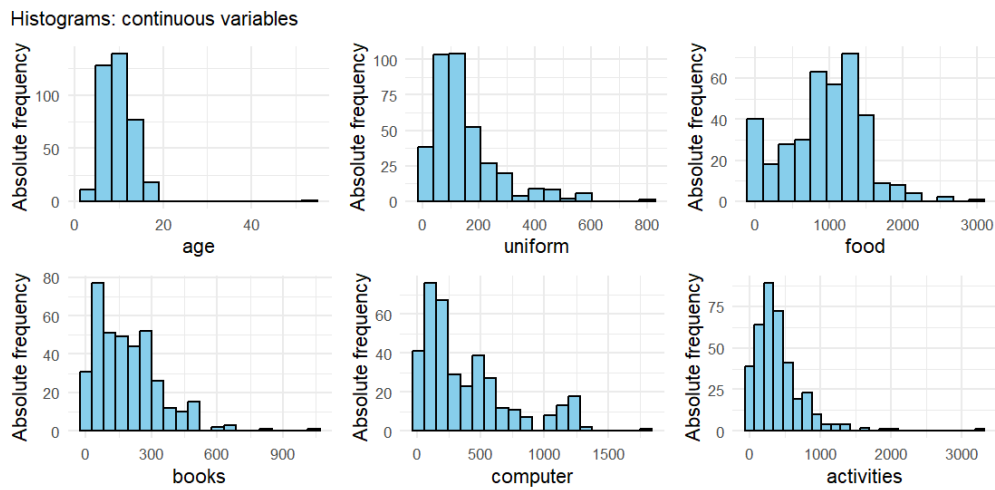


Fig. 2.2. Histograms of the continuous variables EDA

On the other hand, the boxplots of Figure 2.3, which are displayed on a logarithmic scale for better visualization, reveal important insights into the variability of the numeric variables. ‘age’ shows relatively low variability with most observations clustered in a narrow range and few mild outliers, reflecting the expected demographic homogeneity. In contrast, expenditure variables display considerable spread and numerous outliers at the high end, especially ‘food’ and ‘activities’, emphasizing heterogeneity in spending behavior. These findings suggest that transformation methods will be important to handle skewness and outliers in subsequent analyses.

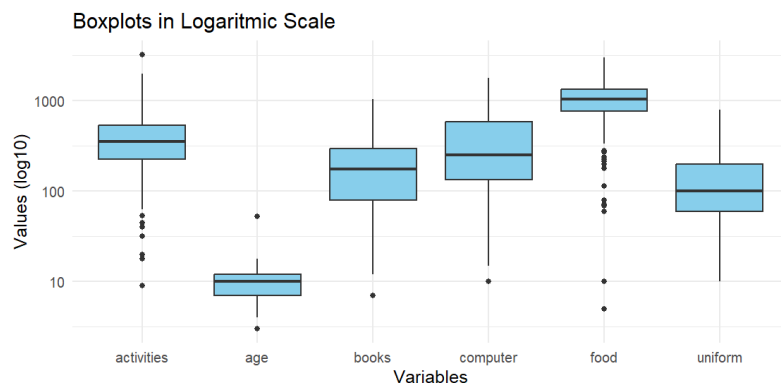


Fig. 2.3. Boxplots of the continuous variables EDA

In summary, this exploratory assessment provides a clear initial understanding of both the categorical and continuous variables in the dataset. In the following section, we turn our attention to the descriptive analysis of the numeric variables in greater detail.

3. DATASET ANALYSIS

3.1. Descriptive analysis

The multivariate analysis first focuses on the quantitative part of the dataset, consisting of the six continuous variables introduced in the previous section. Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ denote the data matrix collecting these variables, where $n = 374$ is the number of individuals and $p = 6$ is the number of quantitative variables. Each row of $\mathbf{X}$ corresponds to one individual, and each column to one of the six variables: ‘age’ (X_1), ‘uniform’ (X_2), ‘food’ (X_3), ‘books’ (X_4), ‘computer’ (X_5), and ‘activities’ (X_6).

An initial overview of the six quantitative variables is provided by the scatterplot matrix in Figure 3.1, which enables a simultaneous examination of their marginal distributions and their pairwise relationships. Along the diagonal, the histograms exhibit strong right skewness for the expenditure variables ‘uniform’, ‘books’, ‘computer’, and ‘activities’: most observations are concentrated near the lower end of their scales, with only a small number of observations taking high values. Notice that the ‘age’ variable (X_1) shows a weaker but still noticeable right tail, while the ‘food’ variable (X_3) displays the broadest range, with most values in a mid-level expenditure range and some reaching up to about 3000 euros. A comparison of these distributions highlights large differences in scale across variables: ‘age’ ranges approximately from 0 to 60, whereas the expenditure variables span several orders of magnitude, with some of them, such as ‘activities’, reaching values close to 4000.

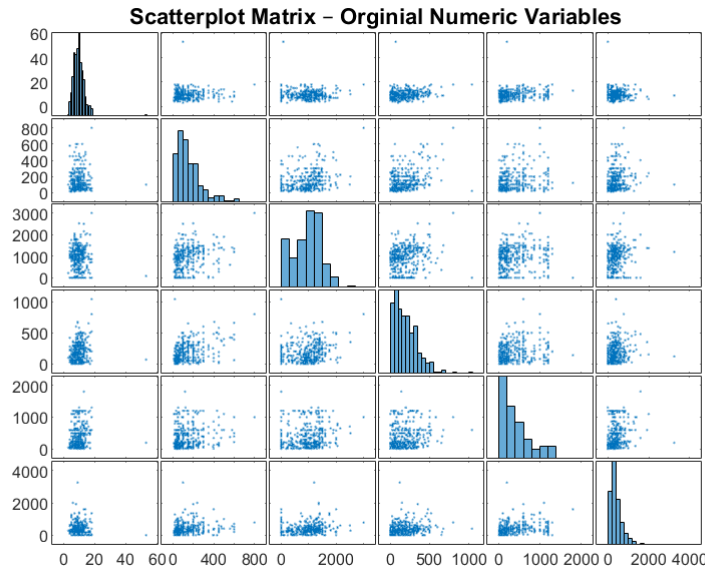


Fig. 3.1. Scatter plot of original data

Furthermore, the off-diagonal scatterplots show that pairwise relationships are generally weak. No pair exhibits a clear linear pattern, and most points are densely grouped

near the lower-left corner, reflecting the high concentration of low expenditures across categories. Only slight upward tendencies appear in a few panels.

Overall, two main issues arise. First, the variables cannot be meaningfully compared due to the large differences in scale. Second, the right-skewness can affect both the correlation structures and any distance-based multivariate analysis, such as principal component analysis.

In addition to the graphical representation of the quantitative data, a multivariate descriptive analysis of the variables is performed to extract further insights. This involves computing the sample mean, median, and variance vectors, which provide a numerical summary of the central tendency and dispersion of the data.

The sample mean vector, $\bar{\mathbf{X}}$, is a multivariate measure of centrality that describes the average position of the observations in the six-dimensional variable space. It can be computed directly from the data matrix as $\bar{\mathbf{X}} = \frac{1}{n}\mathbf{X}'\mathbf{1}$, where $\mathbf{1}$ denotes an $n \times 1$ vector of ones. For our dataset, the sample mean vector is:

```
Sample mean vector =
      9.78      149.52      943.17      191.76      388.49      405.05
```

The first component indicates that the typical student in the data is around 10 years old, indicating that most students are in primary school. The remaining five components describe average annual expenditure across the different categories. These means vary widely in magnitude, which shows that the expenses differ substantially in price and reinforces the scale differences noted in Figure 3.1. Among these categories, ‘food’ expenditure shows the highest mean (943.17€), followed by ‘activities’ (405.05€) and computer-related spending (388.49€). Books and uniforms show much lower average levels (191.76 € and 149.52 €, respectively).

While the mean summarises the typical ‘age’ and spending levels, it can be influenced by outlier values. For this reason, it is also useful to consider the sample median, defined componentwise as the median of each variable:

```
Sample median vector =
      10.00      100.00      995.00      163.00      250.00      330.00
```

The comparison between the mean and median vectors shows that the median values are generally lower than their corresponding means, which is typical of right-skewed distributions. In some variables, the difference is quite large. For instance, ‘computer’ expenditure has a median of 250 € but a mean of 388.49 €, reflecting the impact of a few very high spending values on the mean. For ‘age’ and ‘food’, the median is slightly higher than the mean; however, these differences are very small, and the histograms in Figure 3.1 still display right tails, so it is more appropriate to interpret these variables as mildly right-skewed rather than symmetric or left-skewed.

Beyond central tendency, assessing the dispersion of each variable around its central value provides insight into how spread out the observations are. The resulting sample variance vector for our dataset is:

```

Sample variance vector =
      14.33      15275.24      280748.14      22528.39      122433.42      118758.82

```

These values show that, in general, the variables are highly dispersed and exhibit clear differences in scale. As expected, ‘age’ shows very low variability (14.33), reflecting the narrow ‘age’ range in the sample. In contrast, all expenditure variables have much larger variances, indicating substantial differences in spending levels across individuals. The high variability gaps between variables underscore the strong scale differences in the dataset and justify applying transformations and standardization before using distance-based multivariate methods.

3.1.1. Transformation and standardization of the data

Due to the strong right-skewness and the differences in scale observed in the original variables, a two-step preprocessing procedure is applied. In the first step, a logarithmic transformation $Y_j = \log(X_j + 2)$ is applied to the variables ‘age’ (X_1), ‘uniform’ (X_2), ‘books’ (X_4), ‘computer’ (X_5) and ‘activities’ (X_6). The constant 2 is chosen after checking the minimum values of the variables, ensuring all transformed values remain strictly positive. This transformation reduces the influence of large values and produces more balanced and symmetric marginal distributions.

The ‘food’ variable (X_3), despite being an expenditure variable, behaves differently. Its distribution is concentrated in a mid-range of values, the difference between the mean and the median is relatively small, and the right tail is less extreme. Applying the $\log(x + 2)$ transformation in this case results in a more irregular, slightly left-skewed histogram and produces less regular scatterplots. Since the transformation does not improve either symmetry or the linear structure of the data, ‘food’ is left on its original scale and only included in the subsequent standardization step.

In the second step, all variables (log-transformed except X_3) are centered and standardized. Let $\mathbf{X}^{(*)}$ denote the data matrix after the first step. Centering is performed by applying $\mathbf{X}^{(*)} = \mathbf{H}\mathbf{X}^{(*)}$ where $\mathbf{H} = \mathbf{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}'$ and $\mathbf{1}$ is a vector with all entries 1. The centered matrix is then standardized by dividing each column by its corresponding sample standard deviation, yielding the standardized matrix $\mathbf{Z}$. Once the transformation and the standardization are done, the scatterplot matrix shown in Figure 3.2 is obtained.

The resulting scatterplot matrix exhibits narrower, more symmetric marginal distributions and more ‘uniform’ scatter patterns across variable pairs, indicating that both skewness and scale differences have been reduced.

3.1.2. Covariance and correlation structures

With the standardized data matrix $\mathbf{Z}$ obtained, we can now analyze its multivariate structure through the covariance and correlation matrices. These matrices summarize, respectively, the variability of each standardized variable and the strength of the linear

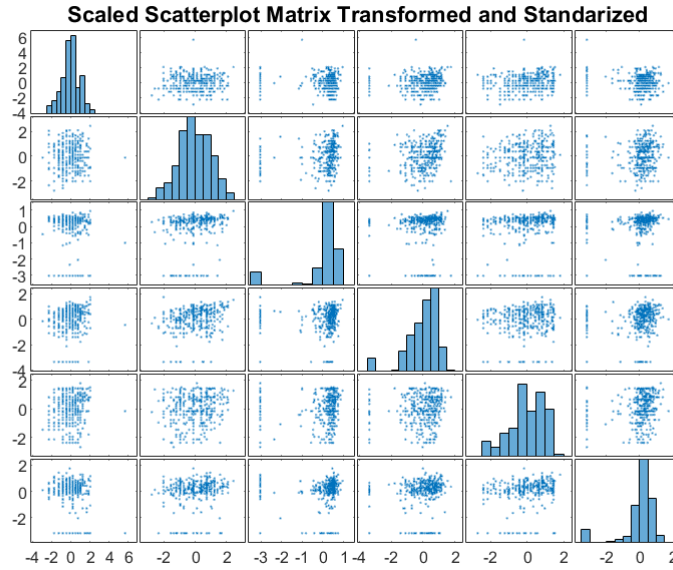


Fig. 3.2. Scatter plot of transformed and standardized data

relationships between all variable pairs.

To quantify these features, we first compute the sample covariance matrix of the standardized data. It is defined as $\mathbf{S} = \frac{1}{n} \mathbf{Z}' \mathbf{H} \mathbf{Z}$ where $\mathbf{H}$ is the centering matrix. On the other hand, the correlation matrix of $\mathbf{Z}$ is defined as the matrix whose elements are the linear correlation coefficients between the columns of $\mathbf{Z}$. Formally, $\mathbf{R} = \mathbf{D}_s^{-1} \mathbf{S} \mathbf{D}_s^{-1}$, where $\mathbf{S}$ is the sample covariance matrix and $\mathbf{D}_s = \text{diag}(\sqrt{s_{11}}, \dots, \sqrt{s_{pp}})$ is the diagonal matrix containing the sample standard deviations of the variables. However, since the data were standardized before computing $\mathbf{S}$, each variable already has variance 1. This implies $\mathbf{D}_s = \mathbf{I}$, and therefore $\mathbf{R} = \mathbf{S}$. In other words, once the variables are centered and scaled to unit variance, the covariance between any two variables coincides with their correlation coefficient. For this reason, the covariance and correlation matrices obtained in our analysis are identical. With our data, the following covariance/correlation matrix is obtained: And it is displayed in the heatmap shown in Figure 3.3.

Correlation matrix R =

1.00	0.04	-0.05	0.10	0.17	-0.16
0.04	1.00	0.28	0.25	0.18	0.15
-0.05	0.28	1.00	0.17	0.13	0.25
0.10	0.25	0.17	1.00	0.15	0.13
0.17	0.18	0.13	0.15	1.00	0.21
-0.16	0.15	0.25	0.13	0.21	1.00

The off-diagonal entries of the correlation matrix summarize the linear relationships between the six standardized variables. Overall, the correlations are weak: most coefficients fall between approximately $-0,05$ and $0,28$. This indicates that the standardized variables do not exhibit strong linear relationships and tend to vary largely independently of one another.

The strongest positive correlation appears between ‘uniform’ expenditure (X_2) and ‘food’ (X_3) with $r \approx 0,28$. This is followed by the correlations between ‘uniform’ (X_2) and

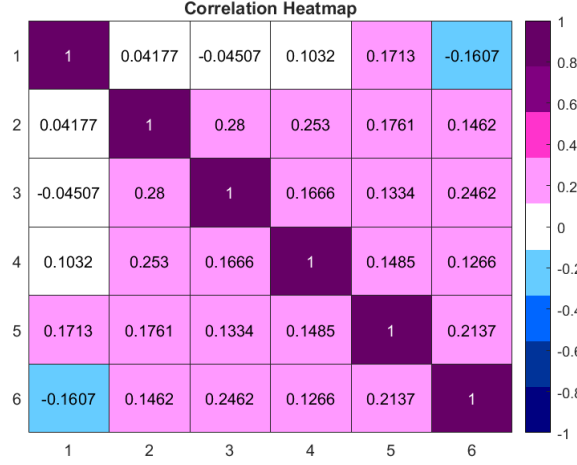


Fig. 3.3. Correlation heatmap.

‘books’ (X_4) and between ‘food’ (X_3) and ‘activities’ (X_6), both around ($r \approx 0,25$). These values indicate weak positive relationships among these expenditure variables. Figure 3.3 highlights these positive correlations through the lighter purple shades in the heatmap.

Negative correlations also appear, but their magnitudes are very small. The largest, approximately $-0,16$, occurs between ‘age’ (X_1) and ‘activities’ (X_6), although this relationship remains weak. These negative values are shown in blue in Figure 3.3. Several variable pairs, such as ‘age’ (X_1) with ‘uniform’ (X_2), ‘food’ (X_3), and ‘books’ (X_4), exhibit correlations extremely close to zero, indicating practical independence. These near-zero correlations appear in white in Figure 3.3.

While pairwise correlations help identify the relationships between specific variable pairs, they do not provide a global view of how strongly the six variables are related as a whole. Rather than examining each coefficient individually, it is often useful to summarize the overall degree of linear dependence in the data using a single intercorrelation measure. This allows us to assess whether the variables, collectively, exhibit a correlation structure or whether they behave independently. Several proposals exist for this purpose, all based on properties of the correlation matrix $\mathbf{R}$, such as its eigenvalues or the diagonal elements of $\mathbf{R}^{-1}$. These give rise to a family of intercorrelation indexes $q_1, \dots, q_6$, each taking values in $[0, 1]$, where values closer to 1 indicate a higher overall amount of intercorrelation among the variables. Computing these six indices for our dataset provides a compact summary of how strongly the variables are related as a whole, complementing the information obtained from the individual correlations.

The six intercorrelation measures computed for our dataset are shown numerically, and plotted in Figure 3.4. Most indices fall between 0,03 and 0,18, indicating very low overall linear dependence among the six variables. Only q_5 reaches a noticeably higher value (approximately 0,58), suggesting that one specific dimension of the correlation structure captures a moderate amount of correlation. However, the remaining indices remain small, reinforcing the earlier conclusion from the correlation matrix that the variables do

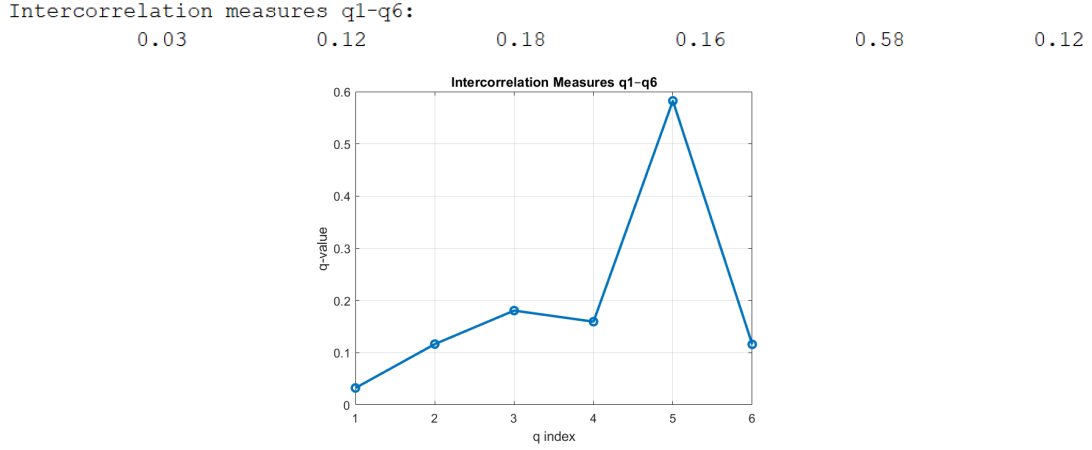


Fig. 3.4. Plot of intercorrelation measures q_1 - q_6

not share a strong common linear structure.

Overall, the intercorrelation measures confirm that the dataset has a weak global correlation structure, with limited redundancy across variables—an observation that is consistent with the low pairwise correlations reported earlier.

For a deeper analysis of the intercorrelation structure, we consider the categorical variables nationality and type of education (each one divided into three groups), and compute the intercorrelation measures separately for each of their three groups.

Figure 3.5 displays the intercorrelation measures $q_1, \dots, q_6$ computed separately for students attending public, state-founded, and private schools. The three profiles share a similar overall pattern, but with notable differences in magnitude. For the first four measures $q_1 - q_4$ and q_6 private schools consistently show the highest values, suggesting a slightly stronger linear dependence among the variables in this group. State-founded schools exhibit intermediate values, while public schools show the lowest intercorrelation. These differences indicate that spending patterns and ‘age’ tend to be more mutually related among students in private schools and less related in public schools. Across all three groups, q_5 is the dominant index, reaching values between 0.61 and 0.68, resembling the behavior observed in the full dataset.

Similarly, Figure 3.6 presents the intercorrelation measures computed separately for students of Spanish nationality, foreign nationality, and double nationality. Here, the differences between groups are more pronounced than in the education-type comparison.

Foreign-nationality students exhibit the highest intercorrelation across almost all indices, indicating a substantially stronger linear structure within this subgroup. Students with double nationality show a similar but slightly weaker pattern. In contrast, the Spanish-nationality group displays very low intercorrelation, suggesting that the variables behave almost independently for these students and showing a pattern similar to that observed in the full dataset. Taken together, these results show that foreign and double-nationality students present a markedly stronger common correlation structure, whereas Spanish stu-

0.02	0.09	0.13	0.12	0.61	0.09
0.08	0.15	0.20	0.15	0.68	0.15
0.18	0.25	0.38	0.23	0.68	0.24

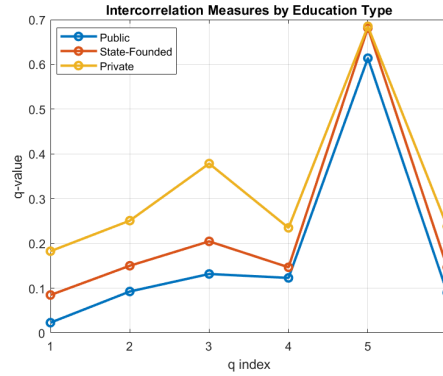


Fig. 3.5. Plot of intercorrelation measures q_1 – q_6 by education

dents exhibit the weakest. This outcome is reasonable given the sample sizes: the Spanish group is much larger (348 observations), while the foreign and double-nationality groups contain only 14 and 12 students, respectively.

0.03	0.12	0.18	0.16	0.58	0.12
0.70	0.62	0.69	0.24	0.92	0.54
0.56	0.49	0.58	0.24	0.87	0.40

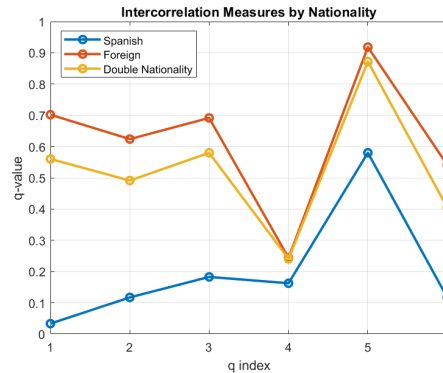


Fig. 3.6. Plot of intercorrelation measures q_1 – q_6 by nationality

Taken together, the intercorrelation measures across nationality and education type show a consistently low degree of overall dependence among the six quantitative variables, matching the overall intercorrelations for the whole dataset.

In summary, the descriptive and intercorrelation analyses reveal that the six quantitative variables exhibit substantial differences in scale, pronounced right-skewness in their original form, and generally weak linear relationships. After applying the appropriate transformations and standardization, the variables lie on a comparable scale, making the dataset well-suited for multivariate techniques. With these preliminary insights established, we now proceed to Principal Component Analysis (PCA) to investigate the main dimensions of variability and extract a more compact representation of the data.

4. PRINCIPAL COMPONENT ANALYSIS (PCA)

4.1. PCA: Brief Theoretical Introduction

Principal Component Analysis (PCA) is a linear technique that transforms a set of possibly correlated variables into a new system of orthogonal components ordered by the amount of variance they explain. Assuming centered data, the first principal component is the direction that maximizes the sample variance, the second captures the largest remaining variance under orthogonality, and so on. Each eigenvalue measures the variance explained by its corresponding component.

Since our variables were previously standardized, all of them are now on the same scale. Therefore, PCA can be computed equivalently from the covariance or the correlation matrix, as both contain the same information after standardization. This prevents high-variance variables from dominating the first component and ensures that all variables contribute comparably. PCA thus helps detect redundancy among variables and, when most variability concentrates in a few components, it allows for an effective reduction of dimensionality.

4.2. PCA computation

After standardising all quantitative variables, we performed the PCA using the covariance matrix $\mathbf{S}$. As stated in Section 3.1.2, since standardization makes all variables comparable in scale, the covariance and correlation matrices contain the same information and thus lead to identical principal components. For this reason, working with $\mathbf{S}$ is equivalent to working with the correlation matrix $\mathbf{R}$ in our setting.

PCA is obtained through the eigen decomposition of $\mathbf{S}$. Let $\lambda_1, \dots, \lambda_p$ be the eigenvalues ordered from largest to smallest, and let $\mathbf{T}$ be the orthogonal matrix whose columns are the corresponding eigenvectors. Each eigenvalue represents the variance explained by its associated principal component, while each column of $\mathbf{T}$ contains the loadings that quantify how the original variables contribute to that component.

The loading matrix returned by MATLAB for our data is:

T = Eigenvectors					
0.06	-0.79	0.09	-0.21	0.34	-0.46
0.49	-0.06	-0.40	-0.27	-0.67	-0.27
0.48	0.24	-0.20	-0.54	0.56	0.28
0.42	-0.23	-0.40	0.73	0.20	0.19
0.41	-0.29	0.66	-0.03	-0.26	0.49
0.42	0.43	0.44	0.25	0.15	-0.60

Fig. 4.1. Orthogonal matrix $\mathbf{T}$ containing the eigenvectors

This matrix summarizes the contribution of each standardized variable to the six principal components. In the next section, we analyze the eigenvalues, the cumulative variance explained, and the graphical representation of the first principal components.

4.2.1. Explained variance and selection of principal components

There are several criteria that can be used to decide how many principal components should be retained in a PCA. Using the MATLAB function `comp2`, we obtained the scree plot and the accumulated explained variance.

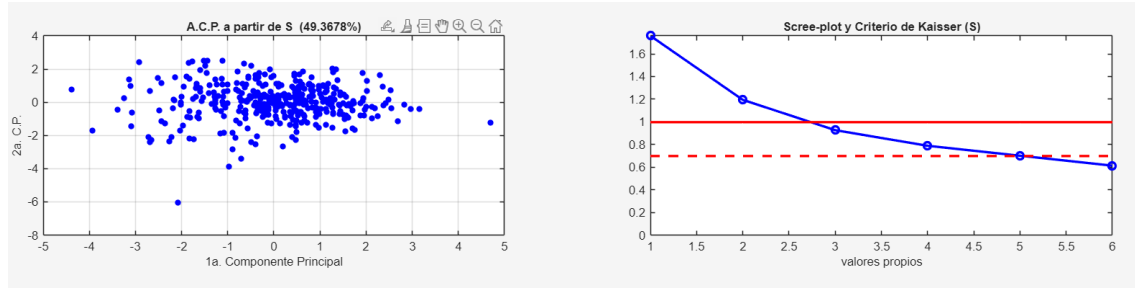


Fig. 4.2. Accumulated explained variance and scree plot

The eigenvalues of the covariance matrix $\mathbf{S}$ are given by:

$$\lambda_S = (1,764456, 1,197613, 0,929173, 0,790939, 0,702916, 0,614902).$$

In Figure 4.2, the left panel shows the projection of all individuals onto the first two principal components. Together, these two components explain **49.37 %** of the total variability, meaning that slightly less than half of the information in the dataset is captured by the plane spanned by the first two axes. Consequently, the two-dimensional representation is limited: although most observations concentrate around the center, a substantial part of the multivariate structure remains in the higher components. This confirms that a bivariate plot is not sufficient to summarize the full complexity of the data.

On the other hand, the scree plot on the right of Figure 4.2 is based on the eigenvalues of the covariance matrix $\mathbf{S}$. The solid red line corresponds to the classical *Kaiser criterion*, which recommends retaining all components with eigenvalues greater than one. According to this rule, the first two components should be selected, since only their eigenvalues satisfy $\lambda_i > 1$:

$$\lambda_1 = 1,7645, \quad \lambda_2 = 1,1976.$$

The dashed red line represents *Jolliffe's criterion*, a more flexible threshold usually set around 0,7. With this rule, the first five components should be retained, as their eigenvalues exceed the 0,7 cutoff:

$$\lambda_1 = 1,7645, \quad \lambda_2 = 1,1976, \quad \lambda_3 = 0,9292, \quad \lambda_4 = 0,7909, \quad \lambda_5 = 0,7029.$$

It is worth noticing that the third eigenvalue,

$$\lambda_3 = 0,9292,$$

is very close to the Kaiser threshold of 1, suggesting that keeping three components may also be a reasonable choice, even though the strict Kaiser rule would select only two. The shape of the scree plot displays a sharp drop from the first to the second component, followed by a more gradual decline, indicating that additional components still retain relevant structure.

The cumulative proportion of variance explained by the successive principal components is

$$(29,41 \%, 49,37 \%, 64,85 \%, 78,04 \%, 89,75 \%, 100 \%).$$

which shows that the first three to four components already account for a substantial share of the total variability.

To complement the graphical criteria, we applied the dimensionality test. For each value of k , the test evaluates the hypothesis

$$H_{0,k} : \lambda_{k+1} = \lambda_{k+2} = \dots = \lambda_p, \quad H_{1,k} : \text{at least one of the last eigenvalues differs.}$$

That is, the test assesses whether the last $p - k$ eigenvalues contain no additional structure and could be discarded. The procedure starts from $k = 1$ and proceeds sequentially, as soon as an $H_{0,k}$ is rejected, the recommended number of retained components is $p - k$.

For our dataset, the test statistics, degrees of freedom, and p -values are given in the following table:

u	df	p-value
1.6549	2	0.4372
5.8501	5	0.3211
17.1510	9	0.0464
50.6547	14	$4,74 \times 10^{-6}$
147.5601	20	$1,84 \times 10^{-21}$

The interpretation is direct. For the first two contrasts ($k = 1$ and $k = 2$), the p -values are relatively large (0,4372 and 0,3211), so the hypotheses

$$H_{0,1} : \lambda_5 = \lambda_6, \quad H_{0,2} : \lambda_4 = \lambda_5 = \lambda_6$$

are not rejected. According to the rule of the test, since $H_{0,3}$ is the first rejected test, the recommended number of retained components is

$$p - k = 6 - 3 = 3.$$

In other words, the dimensionality test suggests keeping three principal components.

The accumulated explained variance provides clear guidance for the final choice of dimensionality. The first two principal components explain only **49.37 %** of the total variability, an insufficient amount for a faithful low-dimensional representation of the dataset. In contrast, the first four components already account for **78.04 %** of the total variance, surpassing the commonly used a threshold of 70 % for an adequate summary of multivariate information.

Considering jointly the behavior of the scree plot, the *Kaiser's* and *Jolliffe's* criteria, and the results of the dimensionality test, retaining four principal components emerges as a balanced and well-supported decision. *Kaiser's* rule suggests keeping two components, while *Jolliffe's* criterion recommends five; the dimensionality test points towards three components. Choosing four components, therefore, lies between these recommendations and ensures that a substantial portion of the information is preserved.

This conclusion is also consistent with the exploratory analysis of the scatterplot matrix, where no strong linear patterns or clusters were observed among the variables. The weak pairwise relationships imply that the original variables capture largely independent aspects of the data, making it reasonable that several principal components are needed to represent the structure adequately.

Overall, retaining the first **four** principal components provides a meaningful reduction of dimensionality, from six original variables to four components, while preserving more than 78 % of the total variability.

4.2.2. Loadings and interpretation of the main principal components

To understand how each variable contributes to the principal components, we examine the loading matrix **T** obtained from MATLAB together with the loadings plot of the first four components shown in Figure 4.3. The loadings, displayed numerically in **T** and visually in the plot, represent the coefficients that define each principal component as a linear combination of the standardized variables.

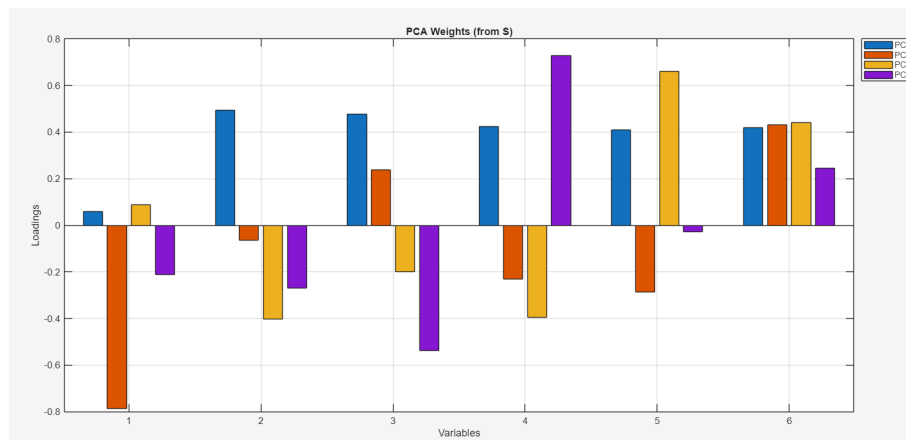


Fig. 4.3. Weights of first four principal components.

Figure 4.3 displays the loadings of the first four principal components, showing how strongly each standardized variable contributes to each component. Positive and negative bars indicate the direction of the contribution, while their height reflects its magnitude. PC1 assigns relatively large positive weights to most expenditure variables, while PC2 shows a contrast between variables with opposite-sign loading. On the other hand, the third and fourth components exhibit more mixed patterns.

The matrix $\mathbf{T}_1$ contains the eigenvectors of the covariance matrix $\mathbf{S}$ and therefore defines the directions of the principal components.

$\mathbf{T}_1 =$

0.06	-0.79	0.09	-0.21	0.34	-0.46
0.49	-0.06	-0.40	-0.27	-0.67	-0.27
0.48	0.24	-0.20	-0.54	0.56	0.28
0.42	-0.23	-0.40	0.73	0.20	0.19
0.41	-0.29	0.66	-0.03	-0.26	0.49
0.42	0.43	0.44	0.25	0.15	-0.60

Fig. 4.4. Principal component matrix.

Using the notation $X_1, \dots, X_6$ for the quantitative variables ('age', 'uniform', 'food', 'books', 'computer', 'activities'), the first principal component is given by:

$$Y_1 = 0,0597 X_1 + 0,4935 X_2 + 0,4769 X_3 + 0,4247 X_4 + 0,4104 X_5 + 0,4205 X_6.$$

As observed in Figure 4.3, all loadings are positive and of comparable magnitude, with the largest contributions coming from X_2 ('uniform'), X_3 ('food'), and X_4 ('books'). Since all coefficients share the same sign, this component reflects a *size dimension*: individuals with high values of Y_1 tend to present globally higher levels of educational expenditure.

In addition, the second principal component is:

$$Y_2 = -0,7867 X_1 - 0,0633 X_2 + 0,2372 X_3 - 0,2314 X_4 - 0,2866 X_5 + 0,4303 X_6.$$

It can be seen that this component shows a mixture of positive and negative loadings, dominated by a strong negative coefficient for 'age' (X_1) and a large positive loading for 'activities' (X_6). Accordingly, Y_2 represents a *shape component*, capturing a contrast mainly between 'age' and participation in educational 'activities'.

The third and fourth principal components exhibit more complex patterns, with several positive and negative loadings interacting across variables. Their interpretation is therefore less direct. As is common in PCA, these components also behave as *shape-type* dimensions, reflecting additional contrasts between different expenditure categories that are not explained by the first two principal directions.

4.3. Sensibility analysis

The stability of the PCA configuration evaluates how much each observation influences the final PCA representation. To measure this effect, we apply a *Jackknife* (leave-one-

out) procedure: for each i , we remove the i -th individual, recompute the PCA, project the deleted observation onto the new configuration, and compare its new coordinates with the original ones. For every individual, we compute the Euclidean distance between its position in the original PCA and its position in each leave-one-out configuration. These distances summarize how sensitive the PCA solution is to the removal of each unit.

Figure 4.5 shows, in a single graphic, both the boxplots of these distances and the 90 % stability regions in the PCA plane. Individuals with larger boxplots or wider ellipses are those whose PCA coordinates change the most, meaning they are less stable and have a higher influence on the PCA configuration. Conversely, smaller boxes and tighter ellipses indicate stable observations. Therefore, we can conclude that variables X_2 and X_4 , associated respectively with uniform and books, are the most stable ones, as they exhibit both short boxplots and compact stability ellipses.

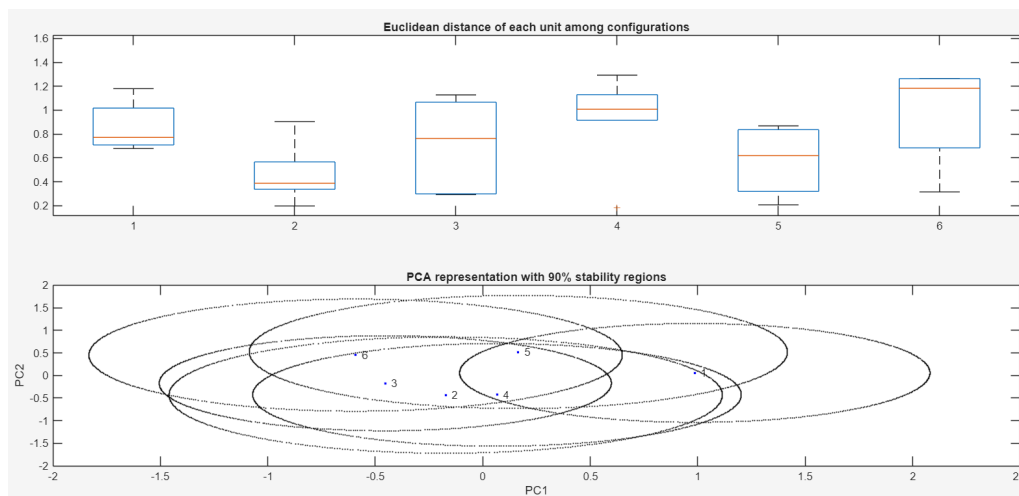


Fig. 4.5. Stability analysis: Euclidean distances across leave-one-out configurations (top) and 90 % stability regions (bottom).

All in all, this study has provided a comprehensive multivariate exploration of the educational expenditure dataset. After the data cleaning process, the exploratory phase revealed substantial differences in scale, pronounced skewness, and weak linear relationships among the quantitative variables. Following the subsequent transformation and standardization steps, the variables became suitable for multivariate analysis. Once the data were properly prepared, PCA allowed us to disentangle these relationships by identifying four principal components that captured more than 78 % of the total variability, effectively summarizing the underlying structure of the data. The loadings revealed a dominant expenditure-related direction, with additional components capturing more specific contrasts among spending categories. The sensitivity analysis further showed that the PCA configuration is stable, since most observations remained consistent under the leave-one-out procedure. Overall, these findings provide a coherent low-dimensional representation of the data and a solid basis for any subsequent statistical analysis.